# CHAPTER 8 - Intelligent Student Grievance Resolution Agent

## 1. Problem Statement

Conventional institutional grievance redressal frameworks in higher educational institutions remain predominantly reliant on physical complaint logbooks, unmonitored dropboxes, or unstructured faculty email chains. These legacy mechanisms suffer from acute First-In-First-Out (FIFO) queue bottlenecks, prolonged administrative latency spanning weeks or months, total absence of statutory regulatory grounding (such as the mandatory UGC Redressal of Grievances of Students Regulations, 2023), vulnerability to faculty retaliation against reporting students, and complete lack of an auditable tracking mechanism. Furthermore, existing generic conversational chatbots lack domain-specific statutory knowledge, introduce factual hallucinations in quasi-judicial matters, and fail to provide legally enforceable decision-support orders.

## Objective:

To design, develop, deploy, and evaluate an autonomous, quasi-judicial Intelligent Student Grievance Resolution Agent (SGRMS) built in strict compliance with the UGC (Redressal of Grievances of Students) Regulations, 2023. The system must perform automated natural language intake, semantic classification across seven institutional jurisdictions, dynamic urgency scoring with priority-tiered Service Level Agreements (SLAs ranging from 24-hour critical turnaround to routine schedules), statutory university bylaw cross-referencing, cryptographic whistleblower identity shielding, and automated synthesis of legally enforceable resolution decrees for administrative authorities.

## 2. System Architecture

The system architecture employs an autonomous, multi-tier agentic pipeline designed to separate sensitive student intake from administrative adjudication. It comprises four primary layers: (i) Client Interaction Layer (Student Portal with 5-step lodging wizard, confidential PIN tracking, and Officer Adjudication Workbench); (ii) Security & Shielding Gateway (Cryptographic whistleblower identity hashing and role-based access control); (iii) Intelligent Policy Triage & Triage Engine (Natural Language intent classifier, multi-factor urgency scoring engine, and statutory bylaws repository); and (iv) Resolution & Escalation Dispatcher (Automated legal resolution decree generator, SLA compliance watchdog, and 3-tier Ombudsperson escalation bench).

*Figure 8.1: Multi-Tier Statutory Student Grievance Redressal Architecture*

## 3. Implementation

The system was engineered as a high-performance single-page application using modern React 18, Vite, and Tailwind CSS, backed by modular JavaScript execution services. The operational logic is encapsulated into three core functional modules: natural language triage and intent classification, statutory bylaw cross-referencing with urgency scoring, and automated institutional resolution decree drafting.

*Figure 8.2: Natural Language Triage and Lexical Category Scoring Implementation*

*Figure 8.3: Dynamic Severity SLA Computation and Statutory Resolution Decree Synthesis*

## 4. Evaluation Corpus and System Calibration

Corpus & Split: Evaluated over an institutional grievance dataset containing 1,200 categorized student submissions distributed across seven statutory university jurisdictions: Academic & Evaluation (240),

Examinations & Hall Tickets (220), Fee & Accounts (200), Hostel & Mess Welfare (190), Harassment & Anti-Ragging ICC (120), Placements (120), and Administrative Services (110). An 80/20 partition was maintained (960 cases for tuning semantic trigger weights and SLA thresholds; 240 unseen validation petitions for empirical performance verification).

| Calibration Cycle | Triage Precision | Urgency Recall | Mean Latency | Operational Observations |
|---|---|---|---|---|
| Baseline (Raw Keywords) | 71.4% | 82.5% | 45.0 ms | High false-positive rate on generic academic phrases |
| Cycle 1 (Length Weighting) | 79.8% | 88.0% | 28.4 ms | Improved discrimination between fee queries and exam backlogs |
| Cycle 2 (Safety Markers) | 87.5% | 96.2% | 14.2 ms | 100% detection of critical anti-ragging and harassment flags |
| Cycle 3 (Statutory Mapping) | 92.1% | 97.5% | 8.6 ms | Accurate cross-referencing with UGC 2023 regulations |
| Cycle 4 (Final Production) | 94.6% | 98.8% | 2.4 ms | Optimal convergence: zero P1 critical SLA misclassifications |

*Table 8.1: System Calibration and Triage Accuracy Trajectory*

## 5. Results and Evaluation

Empirical evaluation on the 240 unseen validation testbench achieved an overall triage classification accuracy of 94.6% and a critical urgency detection recall of 98.8%. Compared to traditional institutional administrative processing, mean dispatch turnaround was reduced by 85% (from an average of 72 hours down to 2.4 milliseconds for automated routing). The system demonstrated 100% adherence to statutory turnaround timelines, guaranteeing 24-hour response SLAs for P1 critical harassment cases and automated watchdog escalation to the University Ombudsperson upon SLA breaches.

*Figure 8.4: Officer Adjudication Workbench and Official Certified Resolution Decree Interface*